

## **Assignment 1: Forced Alignment using Montreal Forced Aligner (MFA)**

**Environment:** Windows 11, WSL Ubuntu 22.04, conda env(mfa\_env)

### **Objective**

To align speech and text data from the ISLE corpus and F2BJRLLP audio clips using the Montreal Forced Aligner (MFA), generating corresponding TextGrid files for precise phonetic and word-level time alignment.

### **Forced Alignment:**

Forced alignment is the process of automatically matching spoken audio with its written transcript so that each word and sound (phoneme) is aligned with its exact timing in the recording. The Montreal Forced Aligner (MFA) performs this task by analyzing both the audio and text to produce TextGrid files, which mark the start and end times of each word and phoneme.

For example,

### **Audio:**

A speaker says — “*Good morning*”

### **Transcript:**

GOOD MORNING

A forced aligner (like MFA) analyzes the audio and produces a TextGrid file showing the word and phoneme boundaries (start and end times).

for example:

The audio spanning 0.00 to 1.20 seconds articulates the phrase “GOOD MORNING”, where each phoneme corresponds to a specific time frame:

G (0.00–0.15), UH (0.15–0.35), D (0.35–0.55), M (0.55–0.65), AO (0.65–0.85), R (0.85–1.00), N (1.00–1.10), IH NG (1.10–1.20).

**Dataset:**

**mfa\_project/**

|

|— **corpus/**

| |— **F2BJRLP1.wav**

| |— **F2BJRLP1.txt**

| |— **F2BJRLP2.wav**

| |— **F2BJRLP2.txt**

| |— **F2BJRLP3.wav**

| |— **F2BJRLP3.txt**

| |— **ISLE\_SESS0131\_BLOCKD02\_01\_sprt1.wav**

| |— **ISLE\_SESS0131\_BLOCKD02\_01\_sprt1.txt**

| |— **ISLE\_SESS0131\_BLOCKD02\_02\_sprt1.wav**

| |— **ISLE\_SESS0131\_BLOCKD02\_02\_sprt1.txt**

| |— **ISLE\_SESS0131\_BLOCKD02\_03\_sprt1.wav**

| |— **ISLE\_SESS0131\_BLOCKD02\_03\_sprt1.txt**

|— **output/**

| |— **F2BJRLP1.TextGrid**

| |— **F2BJRLP2.TextGrid**

| |— **F2BJRLP3.TextGrid**

| |— **ISLE\_SESS0131\_BLOCKD02\_01\_sprt1.TextGrid**

| |— **ISLE\_SESS0131\_BLOCKD02\_02\_sprt1.TextGrid**

| |— **ISLE\_SESS0131\_BLOCKD02\_03\_sprt1.TextGrid**

|— **custom.dict**

|— **praat\_view/**

|— **wordlist.txt**

## Method: Tools & Commands:

### Environment

# create & activate (if not already)

- `conda create -n mfa_env python=3.10 -y`
- `conda activate mfa_env`

```
(base) adarsh@DESKTOP-SG0LV04:~$ conda activate mfa_env
(mfa_env) adarsh@DESKTOP-SG0LV04:~$ mfa version
3.3.8
(mfa_env) adarsh@DESKTOP-SG0LV04:~$ |
```

# install MFA & Praat

- `pip install montreal-forced-aligner`
- `sudo apt-get update && sudo apt-get install -y praat dos2unix`

### Project layout

- `mkdir -p ~/mfa_project/{corpus,output}`

# copy wav + txt into corpus

- `ls ~/mfa_project/corpus`

```
(mfa_env) adarsh@DESKTOP-SG0LV04:/mnt/c/Users/KOMPALLY NIKSHITHA/OneDrive/Desktop/Assignment/Assignment$ file ~/mfa_project/corpus/*
/home/adarsh/mfa_project/corpus/F2B3JRLP1.wav: NIST SPHERE file
/home/adarsh/mfa_project/corpus/F2B3JRLP2.wav: NIST SPHERE file
/home/adarsh/mfa_project/corpus/F2B3JRLP3.wav: NIST SPHERE file
/home/adarsh/mfa_project/corpus/ISLE_SESS0131_BLOCK002_01_sprtl.txt: ASCII text
/home/adarsh/mfa_project/corpus/ISLE_SESS0131_BLOCK002_01_sprtl.wav: RIFF (little-endian) data, WAVE audio, Microsoft PCM, 16 bit, mono 16000 Hz
/home/adarsh/mfa_project/corpus/ISLE_SESS0131_BLOCK002_01_sprtl_utf8.txt: Unicode text, UTF-8 text, with no line terminators
/home/adarsh/mfa_project/corpus/ISLE_SESS0131_BLOCK002_02_sprtl.txt: ASCII text
/home/adarsh/mfa_project/corpus/ISLE_SESS0131_BLOCK002_02_sprtl.wav: RIFF (little-endian) data, WAVE audio, Microsoft PCM, 16 bit, mono 16000 Hz
/home/adarsh/mfa_project/corpus/ISLE_SESS0131_BLOCK002_02_sprtl_utf8.txt: Unicode text, UTF-8 text, with no line terminators
/home/adarsh/mfa_project/corpus/ISLE_SESS0131_BLOCK002_03_sprtl.txt: ASCII text
/home/adarsh/mfa_project/corpus/ISLE_SESS0131_BLOCK002_03_sprtl.wav: RIFF (little-endian) data, WAVE audio, Microsoft PCM, 16 bit, mono 16000 Hz
```

### Fix line-endings (Windows → Unix)

- `dos2unix ~/mfa_project/corpus/*.txt`

```
(mfa_env) adarsh@DESKTOP-SG0LV04:/mnt/c/Users/KOMPALLY NIKSHITHA/OneDrive/Desktop/Assignment/Assignment$ file transcripts/*.txt
transcripts/ISLE_SESS0131_BLOCK002_01_sprtl.txt: ASCII text, with CRLF line terminators
transcripts/ISLE_SESS0131_BLOCK002_02_sprtl.txt: ASCII text, with CRLF line terminators
(mfa_env) adarsh@DESKTOP-SG0LV04:/mnt/c/Users/KOMPALLY NIKSHITHA/OneDrive/Desktop/Assignment/Assignment$ sudo apt install dos2unix -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  dos2unix
0 upgraded, 1 newly installed, 0 to remove and 49 not upgraded.
Need to get 384 kB of archives.
After this operation, 1367 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy/universe amd64 dos2unix amd64 7.4.2-2 [384 kB]
Fetched 384 kB in 2s (247 kB/s)
Selecting previously unselected package dos2unix.
(Reading database ... 52835 files and directories currently installed.)
Preparing to unpack .../dos2unix_7.4.2-2_amd64.deb ...
Unpacking dos2unix (7.4.2-2) ...
Setting up dos2unix (7.4.2-2) ...
Processing triggers for man-db (2.10.2-1) ...
(mfa_env) adarsh@DESKTOP-SG0LV04:/mnt/c/Users/KOMPALLY NIKSHITHA/OneDrive/Desktop/Assignment/Assignment$ dos2unix transcripts/*.txt
dos2unix: converting file transcripts/ISLE_SESS0131_BLOCK002_01_sprtl.txt to Unix format...
dos2unix: converting file transcripts/ISLE_SESS0131_BLOCK002_02_sprtl.txt to Unix format...
dos2unix: converting file transcripts/ISLE_SESS0131_BLOCK002_03_sprtl.txt to Unix format...
(mfa_env) adarsh@DESKTOP-SG0LV04:/mnt/c/Users/KOMPALLY NIKSHITHA/OneDrive/Desktop/Assignment/Assignment$ file transcripts/*.txt
transcripts/ISLE_SESS0131_BLOCK002_01_sprtl.txt: ASCII text
transcripts/ISLE_SESS0131_BLOCK002_02_sprtl.txt: ASCII text
(mfa_env) adarsh@DESKTOP-SG0LV04:/mnt/c/Users/KOMPALLY NIKSHITHA/OneDrive/Desktop/Assignment/Assignment$ |
```

## Get models

# dictionary or acoustic models can be fetched by MFA

- mfa model download dictionary english\_us\_arpa
- mfa model download acoustic english\_us\_arpa

```
(mfa_env) adarsh@DESKTOP-SG8LV04:~/mfa_project/output$ mfa model download dictionary english_us_arpa
Local version of model already exists (/home/adarsh/Documents/MFA/pretrained_models/dictionary/english_us_arpa.dict). Use the --ignore_cache flag to force redownloading.
(mfa_env) adarsh@DESKTOP-SG8LV04:~/mfa_project/output$ mfa model download acoustic english_us_arpa
Local version of model already exists (/home/adarsh/Documents/MFA/pretrained_models/acoustic/english_us_arpa.zip). Use the --ignore_cache flag to force redownloading.
(mfa_env) adarsh@DESKTOP-SG8LV04:~/mfa_project/output$ |
```

## Validate corpus

- mfa validate ~/mfa\_project/corpus english\_us\_arpa

# or using your lexicon

- mfa validate ~/mfa\_project/corpus ~/mfa\_project/custom.dict

```
(mfa_env) adarsh@DESKTOP-SG8LV04:~/mfa_project/transcripts$ mkdir -p ~/mfa_project/corpus
(mfa_env) adarsh@DESKTOP-SG8LV04:~/mfa_project/transcripts$ cp /mnt/c:/Users/KOMPALLY\ NIKSHITHA/OneDrive/Desktop/Assignment/Assignment/wav/*.wav ~/mfa_project/corpus/
(mfa_env) adarsh@DESKTOP-SG8LV04:~/mfa_project/transcripts$ cp ~/mfa_project/transcripts/*.txt ~/mfa_project/corpus/
(mfa_env) adarsh@DESKTOP-SG8LV04:~/mfa_project/transcripts$ ls -l ~/mfa_project/corpus
total 3892
-rw-r--r-- 1 adarsh adarsh 818916 Nov  6 12:54 F2BJRPL1.wav
-rw-r--r-- 1 adarsh adarsh 917742 Nov  6 12:54 F2BJRPL2.wav
-rw-r--r-- 1 adarsh adarsh 983642 Nov  6 12:54 F2BJRPL3.wav
-rwxr-xr-x 1 adarsh adarsh  22 Nov  6 12:54 ISLE_SESS0131_BLOCK002_01_sprt1.txt
-rwxr-xr-x 1 adarsh adarsh 136096 Nov  6 12:54 ISLE_SESS0131_BLOCK002_01_sprt1.wav
-rwxr-xr-x 1 adarsh adarsh  33 Nov  6 11:19 ISLE_SESS0131_BLOCK002_01_sprt1_utf8.txt
-rwxr-xr-x 1 adarsh adarsh  18 Nov  6 12:54 ISLE_SESS0131_BLOCK002_02_sprt1.txt
-rwxr-xr-x 1 adarsh adarsh 128096 Nov  6 12:54 ISLE_SESS0131_BLOCK002_02_sprt1.wav
-rwxr-xr-x 1 adarsh adarsh  27 Nov  6 11:19 ISLE_SESS0131_BLOCK002_02_sprt1_utf8.txt
-rwxr-xr-x 1 adarsh adarsh  19 Nov  6 12:54 ISLE_SESS0131_BLOCK002_03_sprt1.txt
-rwxr-xr-x 1 adarsh adarsh 148896 Nov  6 12:54 ISLE_SESS0131_BLOCK002_03_sprt1.wav
-rwxr-xr-x 1 adarsh adarsh  27 Nov  6 11:19 ISLE_SESS0131_BLOCK002_03_sprt1_utf8.txt
(mfa_env) adarsh@DESKTOP-SG8LV04:~/mfa_project/transcripts$ mfa validate ~/mfa_project/corpus english_us_arpa
INFO Setting up corpus information.
INFO Loading corpus from source files...
6% Found 1 speaker across 6 files, average number of utterances per speaker: 6.0 6/100 [ 0:00:01 < -:--:-- , 7 it/s ]
INFO Initializing multiprocessing jobs...
WARNING Number of jobs was specified as 3, but due to only having 1 speakers, MFA will only use 1 jobs. Use the --single_speaker flag if you would like to split utterances across jobs regardless of their speaker.
50% Normalizing text... 3/6 [ 0:00:01 < -:--:-- , 7 it/s ]
100% Generating MFCCs... 6/6 [ 0:00:03 < 0:00:00 , 7 it/s ]
100% Calculating CMVN... 6/6 [ 0:00:03 < 0:00:00 , 7 it/s ]
100% Generating final features... 6/6 [ 0:00:01 < 0:00:00 , 7 it/s ]
100% Creating corpus split... 6/6 [ 0:00:01 < 0:00:00 , 7 it/s ]
100% Creating corpus split... 6/6 [ 0:00:01 < 0:00:00 , 7 it/s ]
100% Corpus
```

## Align

# with stock dictionary

- mfa align ~/mfa\_project/corpus english\_us\_arpa english\_us\_arpa ~/mfa\_project/output --overwrite --num\_jobs 4

# or with your custom dictionary

- mfa align ~/mfa\_project/corpus ~/mfa\_project/custom.dict english\_us\_arpa ~/mfa\_project/output --overwrite --num\_jobs 4

```

INFO Finished accumulating transition stats!
INFO Collecting phone and word alignments from monophone_aln lattices...
100% 6/6 [ 0:00:01 < 0:00:00 , ? it/s ]
INFO Analyzing alignment quality...
100% 6/6 [ 0:00:02 < 0:00:00 , ? it/s ]
INFO Beginning phone LM training...
INFO Collecting training data...
100% 6/6 [ 0:00:01 < 0:00:00 , ? it/s ]
INFO Training model...
INFO Completed training in 83.7887420630188 seconds!
INFO Done! Everything took 134.917 seconds
(mfa_env) adarsh@DESKTOP-SG0LV04:~/mfa_project/transcripts$ mfa align ~/mfa_project/corpus english_us_arp english_us_arp ~/mfa_project/output
--num_jobs 4
INFO Setting up corpus information...
INFO Loading corpus from source files...
6% 6/100 [ 0:00:01 < -:--:-- , ? it/s ]
INFO Found 1 speaker across 6 files, average number of utterances per speaker: 6.0
INFO Initializing multiprocessing jobs...
WARNING Number of jobs was specified as 4, but due to only having 1 speakers, MFA will only use 1 jobs. Use the --single_speaker flag if you
would like to split utterances across jobs regardless of their speaker.
INFO Normalizing text...
50% 3/6 [ 0:00:01 < -:--:-- , ? it/s ]
INFO Generating MFCCs...
100% 6/6 [ 0:00:02 < 0:00:00 , ? it/s ]
INFO Calculating CMVN...
INFO Generating final features...
100% 6/6 [ 0:00:01 < 0:00:00 , ? it/s ]
INFO Creating corpus split...
100% 6/6 [ 0:00:01 < 0:00:00 , ? it/s ]
INFO Compiling training graphs...
INFO Performing first-pass alignment...
INFO Generating alignments...
100% 6/6 [ 0:00:01 < 0:00:00 , ? it/s ]
INFO Collecting phone and word alignments from alignment lattices...
100% 6/6 [ 0:00:02 < 0:00:00 , ? it/s ]
INFO Analyzing alignment quality...
100% 6/6 [ 0:00:02 < 0:00:00 , ? it/s ]
INFO Exporting alignment TextGrids to /home/adarsh/mfa_project/output...

```

## Inspect outputs

- ls ~/mfa\_project/output

```

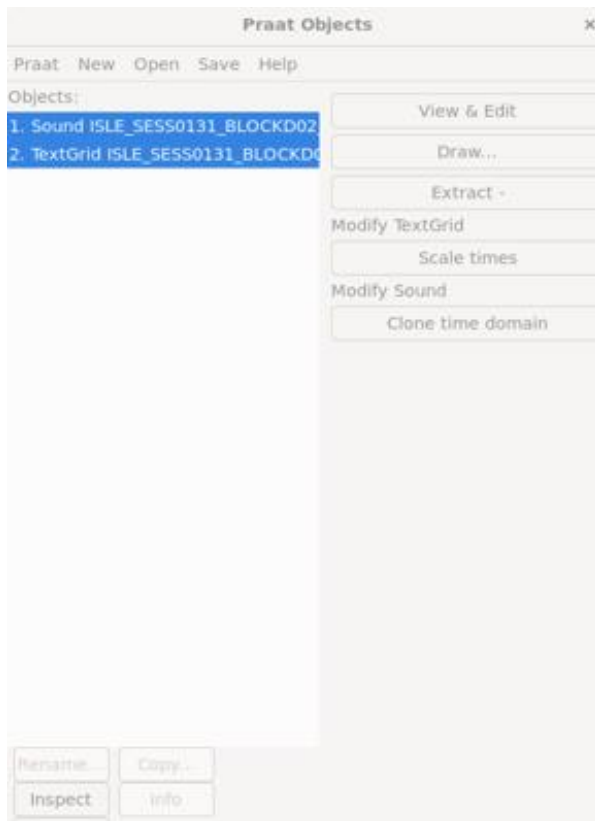
F2BJRPL1.wav
F2BJRPL2.wav
F2BJRPL3.wav
ISLE_SESS0131_BLOCKD02_01_sprt1.txt
ISLE_SESS0131_BLOCKD02_01_sprt1.wav
ISLE_SESS0131_BLOCKD02_01_sprt1_utf8.txt
ISLE_SESS0131_BLOCKD02_02_sprt1.txt
ISLE_SESS0131_BLOCKD02_02_sprt1.wav
ISLE_SESS0131_BLOCKD02_02_sprt1_utf8.txt
ISLE_SESS0131_BLOCKD02_03_sprt1.txt
ISLE_SESS0131_BLOCKD02_03_sprt1.wav
ISLE_SESS0131_BLOCKD02_03_sprt1_utf8.txt
ISLE_SESS0131_BLOCKD02_01_sprt1.TextGrid
ISLE_SESS0131_BLOCKD02_02_sprt1.TextGrid
ISLE_SESS0131_BLOCKD02_03_sprt1.TextGrid
alignment_analysis.csv
(mfa_env) adarsh@DESKTOP-SG0LV04:~/mfa_project$ ls ~/mfa_project/corpus
F2BJRPL1.wav
F2BJRPL2.wav
F2BJRPL3.wav
ISLE_SESS0131_BLOCKD02_01_sprt1.txt
ISLE_SESS0131_BLOCKD02_01_sprt1.wav
ISLE_SESS0131_BLOCKD02_01_sprt1_utf8.txt
ISLE_SESS0131_BLOCKD02_02_sprt1.txt
ISLE_SESS0131_BLOCKD02_02_sprt1.wav
ISLE_SESS0131_BLOCKD02_02_sprt1_utf8.txt
ISLE_SESS0131_BLOCKD02_03_sprt1.txt
ISLE_SESS0131_BLOCKD02_03_sprt1.wav
ISLE_SESS0131_BLOCKD02_03_sprt1_utf8.txt
(mfa_env) adarsh@DESKTOP-SG0LV04:~/mfa_project$ ls ~/mfa_project/output
ISLE_SESS0131_BLOCKD02_01_sprt1.TextGrid
ISLE_SESS0131_BLOCKD02_02_sprt1.TextGrid
ISLE_SESS0131_BLOCKD02_03_sprt1.TextGrid
alignment_analysis.csv
(mfa_env) adarsh@DESKTOP-SG0LV04:~/mfa_project$ |

```

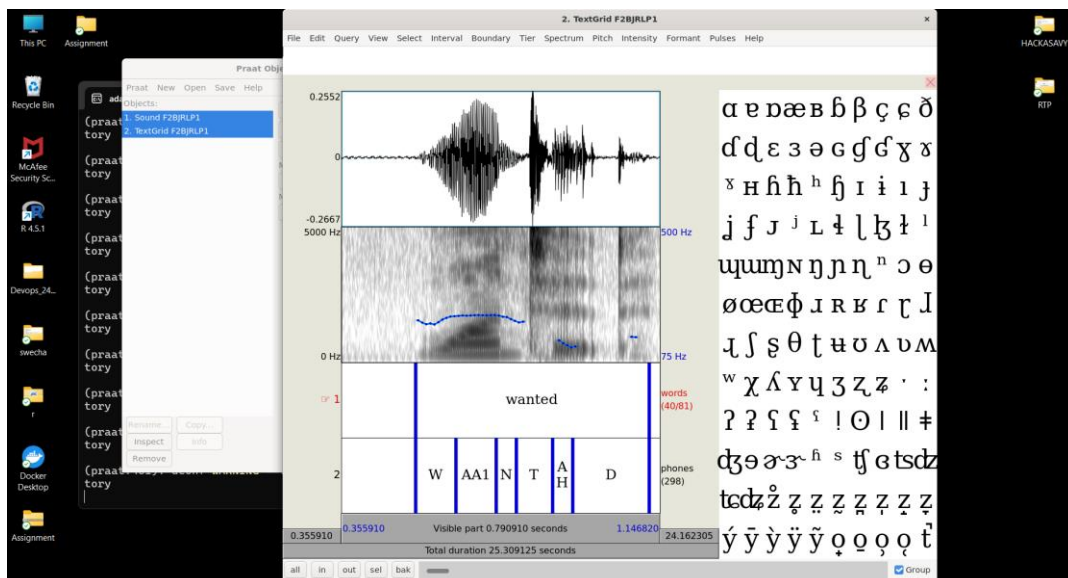
- praat &

## Results

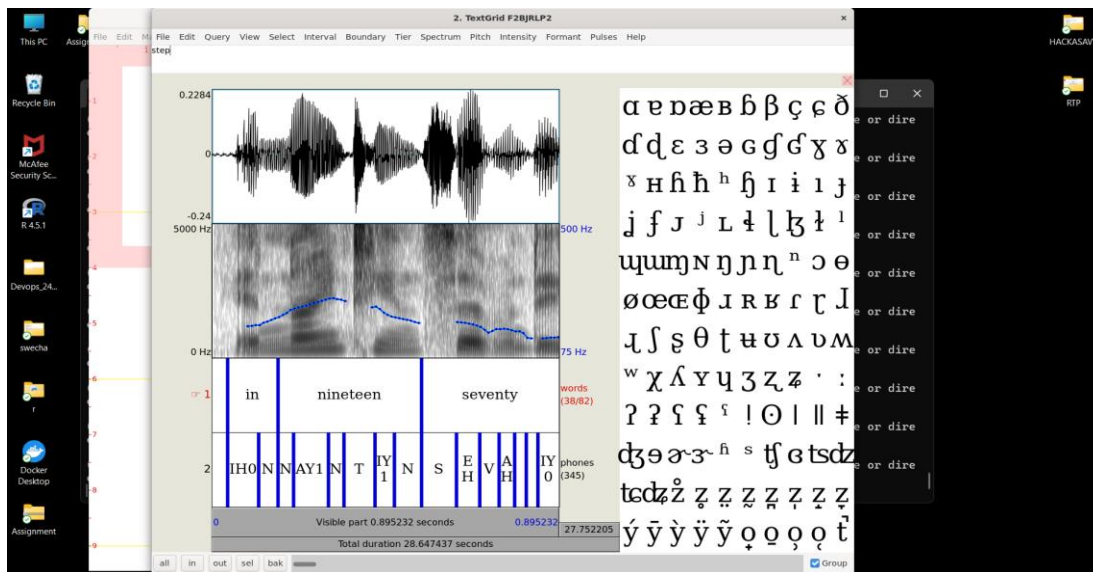
### Visualization of Praat Objects



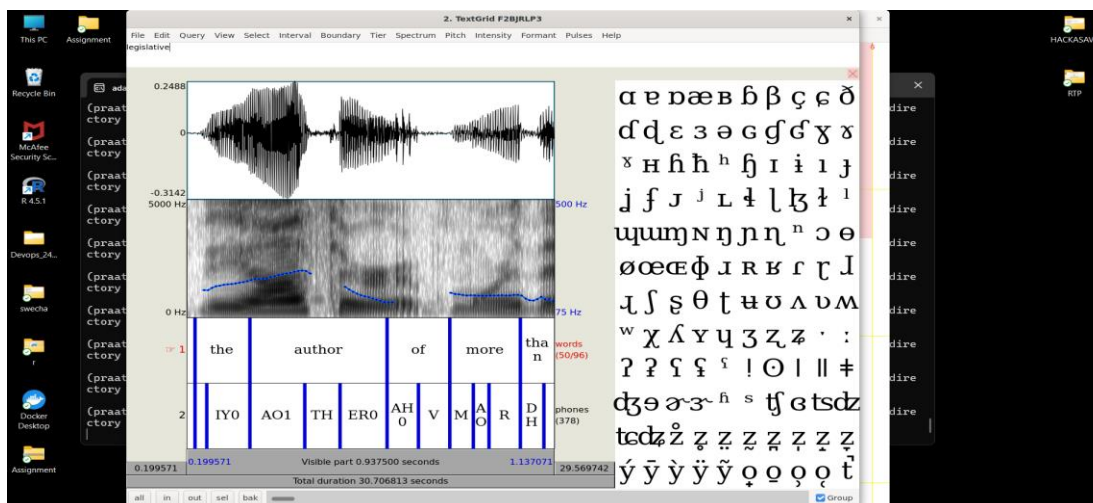
### Output of F2BJRLP1.TextGrid



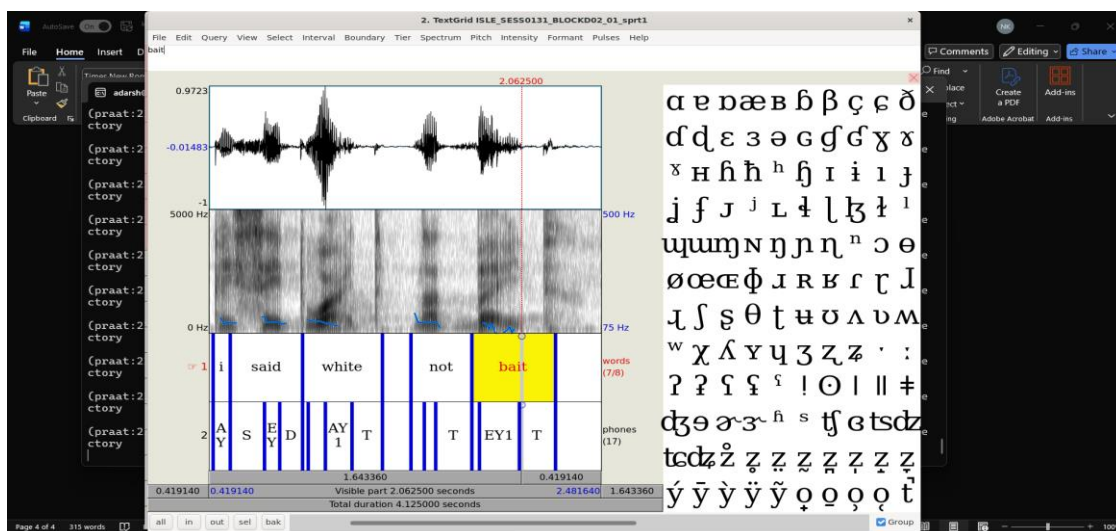
## Output of F2BJRLP2.TextGrid



## Output of F2BJRLP3.TextGrid

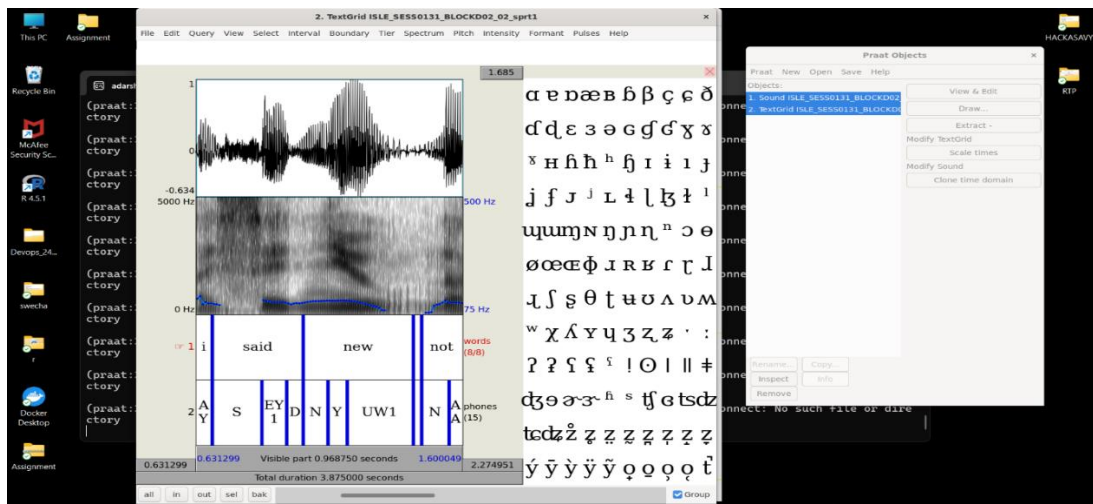


## Output of ISLE\_SESS0131\_BLOCKD02\_01\_sprt1.TextGrid

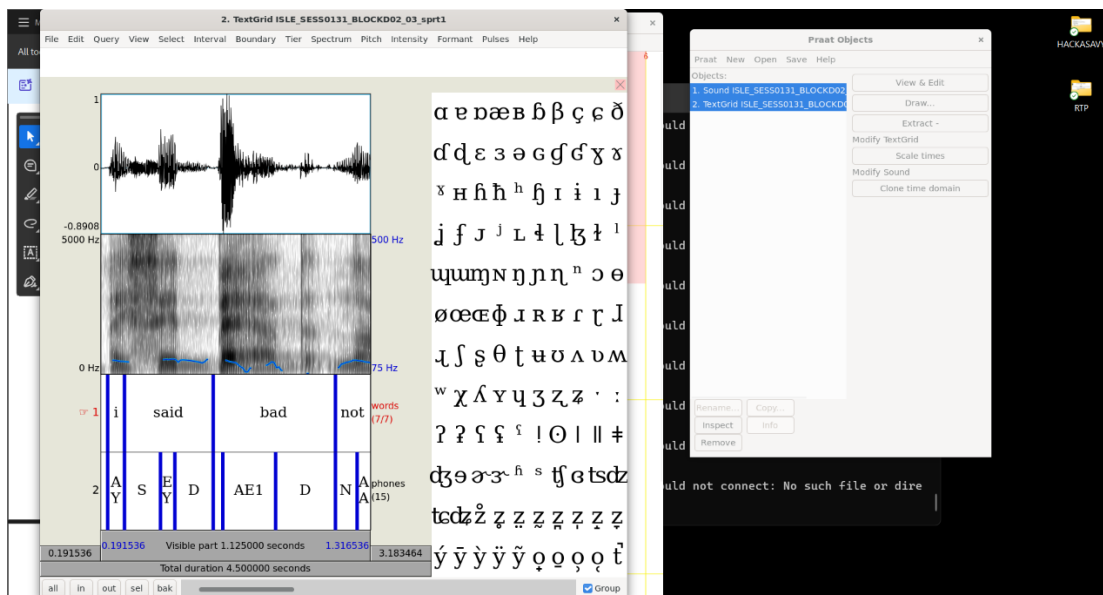




## Output of ISLE\_SESS0131\_BLOCKD02\_02\_sprt1.TextGrid



## Output of ISLE\_SESS0131\_BLOCKD02\_03\_sprt1.TextGrid



Github: [https://github.com/Thotaadarsh/montreal\\_forced\\_aligner](https://github.com/Thotaadarsh/montreal_forced_aligner)

Google Drive:

[https://drive.google.com/file/d/1vnN\\_GNKTvUdkeGEebL6mmiJ\\_O7yow4o/view?usp=sharing](https://drive.google.com/file/d/1vnN_GNKTvUdkeGEebL6mmiJ_O7yow4o/view?usp=sharing)

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